FLEETFLOW ENTERPRISE SPECIFICATION

Frontend Technical Reference

Comprehensive Source Documentation, Function Line Numbers & Stacking Architecture

DOCUMENT METADATA

Version: 1.0.0 (Production Verified Baseline)

Date of Verification: October 11, 2026

Architecture: Vanilla JS SPA + Express REST API + MongoDB Atlas

Security Status: Verified Zero Secrets Leakage (Passwords & URIs Redacted)

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1. FRONTEND FILE INVENTORY

The FleetFlow frontend is implemented with Vanilla JavaScript, HTML5, and CSS3, maintaining complete framework independence for high-performance rendering and lightweight deployment.

File Path	Size	Lines	Role & Primary Responsibilities	Dependencies
index.html	19.4 KB	382	Single-page authenticated application shell with navigation, views & modals	style.css, Leaflet.css, script.js
login.html	22.0 KB	433	Split-screen enterprise login & organization registration view with dual tabs	Embedded CSS & Vanilla JS
script.js	74.2 KB	1853	Full client-side application logic, state, REST client, Leaflet map, DOM bindings	index.html, Leaflet.js, OSRM API

File Path	Size	Lines	Role & Primary Responsibilities	Dependencies
sstyle.css	20.9 KB	1251	Complete design system: tokens, typography, layouts, tables, modals, isolation	CSS Custom Properties (:root)
style.css	20.9 KB	1251	Synchronized release stylesheet for production deployment	Identical to sstyle.css
vendor/leaflet/*	145 KB	N/A	Open-source Leaflet map rendering engine and stylesheet for road trajectories	OpenStreetMap, OSRM

2. HTML STRUCTURE & COMPONENT RANGES

The HTML document index.html is organized into 16 distinct sections ensuring semantic hierarchy, accessibility (ARIA), and strict separation of UI components.

Section Banner	Element Selector / ID	Source Lines	Description
SECTION 1	<head> (Lines 1-9)	1 – 9	Document metadata, viewport settings, title tag
SECTION 2	<link rel="stylesheet">	10 – 14	Vendor Leaflet CSS and core application style.css
SECTION 3		15 – 22	Application grid wrapper and sidebar backdrop element
SECTION 4	<aside class="sidebar">	23 – 79	Navigation drawer with logo, brand mark, 7 nav tabs, and footer
SECTION 5	<header class="topbar">	80 – 130	Header with menu-toggle, search bar, notify bell, and user chip
SECTION 6	Authentication Handlers	131 – 134	Session validation checks redirecting unauthenticated users
SECTION 7	<section id="page-dashboard">	135 – 165	Fleet Command Center: KPI cards, Fuel chart, Donut chart, Activity feed
SECTION 8	<section id="page-vehicles">	166 – 200	Vehicle inventory table, search, status filter, and + Add Vehicle
SECTION 9	<section id="page-drivers">	201 – 232	Admin driver directory, chip filters, safety scores, and + Add Driver
SECTION 10	<section id="page-users">	325 – 360	Employee management directory with role filters and account toggles
SECTION 11	<section id="page-fuel">	233 – 262	Fuel purchase history table, vehicle filter, date filter, and + Add Fuel
SECTION 12	<section id="page-maintenance">	263 – 272	Kanban maintenance board with Overdue, Upcoming, Completed columns
SECTION 13	<section id="page-routes">	273 – 324	Interactive Leaflet road map container, metrics panel, and trips table
SECTION 14		361 – 372	Modal dialog container with dynamic form injection body
SECTION 15		373 – 375	Fixed toast notification container for operational feedback messages
SECTION 16	<script> Tags	376 – 382	Leaflet.js library script followed by core script.js

3. CSS REFERENCE, STACKING CONTEXTS & ISOLATION GUIDE

The design tokens and styles in sstyle.css are organized into 19 distinct functional modules using CSS custom properties and strict stacking isolation:

CSS Section	Key Selectors	Source Lines	Styling Purpose & Stacking Order
Section 1: Design Tokens	:root, --sidebar, --primary, --green	1 – 28	Global theme tokens: slate navbar, indigo primary, radius, shadows
Section 2: Reset & Base	*, *::before, html, body	29 – 45	Box sizing, full height reset, background and body defaults
Section 3: Typography	body, button, input, select, svg	46 – 68	Segoe UI system typography, inherit font rules, svg fills
Section 4: Buttons & Forms	.action-bar, .btn, .btn-primary, .chip	614 – 694	Inputs, buttons with hover lifts, filter chips, action toolbars

CSS Section	Key Selectors	Source Lines	Styling Purpose & Stacking Order
Section 5: Shared Layout	.app, .shell, .content, .page	70 – 76, 404 – 415	Full-height flex app shell, main content padding, page toggling
Section 6: Sidebar & Nav	.sidebar, .brand, .side-nav, .nav-item	77 – 187	Sticky 260px sidebar (z-index: 40), active tab pill highlight
Section 7: Header & Controls	.topbar, .search-wrap, .panel, .user-chip	188 – 403	Sticky topbar (z-index: 30), popovers (.panel, .user-menu z-index: 50)
Section 8: Login Integration	.content, .page.is-visible	404 – 415	Content visibility switches; login split-screen in login.html
Section 9: Dashboard Styles	.kpi-grid, .bar-chart, .donut, .activity	416 – 613	KPI stat cards, 7-day bar chart columns, conic donut, event feed
Section 10-13: CRUD Styles	.action-bar, table, .badge	614 – 761	Shared tabular views for vehicles, drivers, fuel, users
Section 14: Maintenance	.kanban, .kanban-col, .ticket	762 – 815	3-column Kanban layout with color-coded ticket borders
Section 15: Routes & Map Isolation	.map-card, .route-map-container, leaflet	816 – 935	Leaflet map card. isolation: isolate; z-index: 1; traps map controls
Section 16: Tables & Badges	.table-wrap, table, th, td, .badge	695 – 761	Responsive tables, sticky headers, rounded pill status badges
Section 17: Modals & Toasts	.modal-overlay, .modal, .form-grid, .toast	936 – 1094	Modal backdrop (z-index: 80), toast notifications (z-index: 100)
Section 18: Responsive Layout	@media (max-width: 1200px / 840px / 760px)	1095 – 1196	Grid collapsing, icon-only sidebar under 840px, drawer under 760px
Section 19: Accessibility	.sr-only, :focus-visible, [hidden], RBAC	1197 – 1251	Screen-reader classes, high-contrast focus rings, role hiding

!9þ Resolved Stacking Context Hierarchy

Root Stacking Order: Map Canvas (z-index: 1, isolation: isolate) < Sticky Header .topbar (z-index: 30) < Sidebar Backdrop (z-index: 35) < Sidebar (z-index: 40) < Profile Menu Dropdown (z-index: 50) < Modals (z-index: 80) < Toast Alerts (z-index: 100). Leaflet controls (z-index 1000) are cleanly trapped inside the container by CSS isolation.

4. JAVASCRIPT FUNCTION INDEX (SCRIPT.JS)

Below is the complete function inventory extracted directly from the optimized script.js. Every function body start and end line has been verified against the active source:

Function Name	File	Start	End	Purpose	Called By
checkAuthAndInit	script.js	68	100	Verify JWT session cookie via /api/auth/me and initiate dashboard loading	DOMContentLoaded
applyEmployeeNavLabels	script.js	106	123	Customize navigation labels and page descriptions for Employee role	checkAuthAndInit
applyAdminNavLabels	script.js	125	144	Apply full administrative navigation labels and copy	checkAuthAndInit
updateUserChip	script.js	146	161	Render user initials, full name and role badge in top navigation bar	checkAuthAndInit, submitProfile
loadAllFromBackend	script.js	167	211	Fetch all operational datasets via REST APIs (scoped by user role & tenant)	checkAuthAndInit, CRUD submits
renderAll	script.js	213	226	Dispatch rendering pipeline across all application tabs and views	loadAllFromBackend
bindEvents	script.js	232	382	Register all DOM click, input, filter and form event listeners	checkAuthAndInit
handleLogout	script.js	384	394	Terminate authentication session via POST /api/auth/logout	user menu logout button
renderNotifications	script.js	396	430	Compute alert counts for overdue tickets & offline vehicles; render notify popover	renderAll, bindEvents
showSection	script.js	436	470	Switch active tab, toggle DOM page visibility, update topbar header title	nav buttons, search clicks

Function Name	File	Start	End	Purpose	Called By
toggleSidebar	script.js	472	481	Toggle mobile slide-out navigation drawer and backdrop	menu-toggle button
closeSidebar	script.js	483	491	Dismiss mobile navigation drawer	backdrop click, nav clicks
togglePanel	script.js	493	501	Toggle dropdown popovers (notifications, user profile menu)	topbar icon buttons
closePopovers	script.js	503	518	Close all active dropdown popovers and search flyouts	global document clicks
getDriverById	script.js	524	526	Lookup driver object in state cache by unique string ID	getDriverName, vehicle renderers
getVehicleById	script.js	528	530	Lookup vehicle object in state cache by unique string ID	getVehicleName, trip renderers
getDriverName	script.js	532	535	Resolve driver display name or return 'Unassigned'	renderVehicles, renderFuel, renderRoutes
getVehicleName	script.js	537	540	Format vehicle identifier and make/model string	renderTrips, selectTripRoute
formatNumber	script.js	542	544	Format integer with localized thousand separators (en-IN)	renderers, metrics
formatMoney	script.js	546	548	Format rupee monetary string with INR symbol ([1])	renderDashboard, renderFuel, renderMaintenance
statusBadge	script.js	550	572	Generate colored HTML status badge pill element	table renderers, popups
escapeHtml	script.js	574	581	Sanitize user-provided strings against Cross-Site Scripting (XSS)	all HTML template generators
renderDashboard	script.js	587	614	Render KPI cards, fuel consumption bar chart, donut chart, and activity stream	renderAll
kpiCard	script.js	616	621	HTML component generator for top metric KPI summary cards	renderDashboard
monthlyFuelSpend	script.js	623	627	Calculate current month total fuel expenditure	renderDashboard
renderFuelChart	script.js	629	649	Render weekly 7-day fuel consumption bar graph	renderDashboard
renderStatusDonut	script.js	651	674	Frontend operational utility function	Internal module caller
legendItem	script.js	676	679	HTML component for chart legends	renderDonut
renderActivity	script.js	681	689	Render operational event stream in recent activity feed	renderDashboard
renderVehicles	script.js	695	726	Filter and render vehicle inventory table rows and view actions	renderAll, filter inputs
emptyRow	script.js	728	730	Generate empty state table row spanning all columns	table renderers
renderDrivers	script.js	736	769	Render driver roster with license, safety score and reassignment links	renderAll, filter chips
calculateFleetAverage	script.js	775	779	Compute mean km/L fuel efficiency across all recorded fuel logs	renderFuel
calculateFuelEfficiency	script.js	781	791	Calculate incremental km/L efficiency between consecutive fill-ups	renderFuel
renderFuelFilters	script.js	793	802	Populate vehicle filter dropdown options in fuel view	renderAll
renderFuel	script.js	804	838	Filter and render fuel log records table with efficiency indicators	renderAll, filter inputs
classifyMaintenance	script.js	844	849	Categorize maintenance tickets into overdue, upcoming, or completed	renderMaintenance
renderMaintenance	script.js	851	872	Render kanban board columns for fleet maintenance lifecycle	renderAll
kanbanColumn	script.js	874	889	HTML generator for individual maintenance kanban status columns	renderMaintenance
toggleTicketStatus	script.js	891	909	Send PATCH request to mark maintenance ticket Completed	kanban click handler
renderRoutes	script.js	915	965	Render trip dispatch log table and road corridor details	renderAll

Function Name	File	Start	End	Purpose	Called By
initOrUpdateRouteMap	script.js	974	1005	Initialize Leaflet.js interactive map container and OpenStreetMap tile layer	showSection, renderRoutes
selectTripRoute	script.js	1007	1181	Fetch OSRM highway geometry and render driving trajectory on map	renderRoutes, trip row clicks
updateTripStatus	script.js	1183	1203	Send PATCH request to update trip status (On Time, Delayed, etc.)	route status select change
renderUsers	script.js	1209	1245	Render employee accounts table with role and status toggles	renderAll, user filter changes
handleToggleUserStatus	script.js	1247	1265	Send PATCH request to activate or deactivate employee login account	user table action buttons
runGlobalSearch	script.js	1271	1324	Perform cross-entity search across vehicles, drivers and routes	global-search input
openModal	script.js	1330	1402	Configure and display modal dialog with specific form template	add buttons, view actions
closeModal	script.js	1404	1412	Hide active modal dialog and clear form content	modal close button, cancel buttons
optionList	script.js	1414	1419	Generate HTML option elements for select dropdowns	form generators
field	script.js	1421	1428	Generate form field container with label, input and validation error slot	form generators
selectField	script.js	1430	1435	Frontend operational utility function	Internal module caller
formActions	script.js	1437	1440	Frontend operational utility function	Internal module caller
vehicleFormHtml	script.js	1442	1467	Generate Add Vehicle modal form markup	openModal('add-vehicle')
driverFormHtml	script.js	1469	1484	Generate Add Driver modal form markup with decoupled driver profile	openModal('add-driver')
fuelFormHtml	script.js	1486	1498	Generate Add Fuel Log modal form markup	openModal('add-fuel')
maintenanceFormHtml	script.js	1500	1528	Generate Schedule Service modal form markup	openModal('add-maintenance')
reassignFormHtml	script.js	1530	1545	Generate Driver Vehicle Reassignment modal form markup	openModal('reassign')
userFormHtml	script.js	1547	1571	Generate Create Employee Account modal form markup	openModal('add-user')
profileFormHtml	script.js	1573	1583	Generate User Profile update modal form markup	openModal('profile')
vehicleDetailsHtml	script.js	1586	1607	Generate detailed vehicle specification and trip history dialog	openModal('vehicle-details')
detail	script.js	1609	1611	Generate definition list key-value item for vehicle details	vehicleDetailsHtml
handleModalSubmit	script.js	1617	1639	Route form submit event to corresponding entity creation handler	modal form submit
setFieldError	script.js	1641	1646	Frontend operational utility function	Internal module caller
clearErrors	script.js	1648	1656	Frontend operational utility function	Internal module caller
required	script.js	1658	1670	Frontend operational utility function	Internal module caller
submitVehicle	script.js	1672	1704	Validate and POST new vehicle to /api/vehicles in MongoDB	handleModalSubmit
submitDriver	script.js	1706	1740	Validate and POST new decoupled driver profile to /api/drivers	handleModalSubmit
submitFuel	script.js	1742	1777	Validate and POST fuel log to /api/fuel in MongoDB	handleModalSubmit
submitMaintenance	script.js	1779	1806	Validate and POST maintenance record to /api/maintenance	handleModalSubmit
submitReassign	script.js	1808	1827	Validate and PUT driver vehicle reassignment to /api/drivers/:id/assign	handleModalSubmit
submitUser	script.js	1829	1890	Validate and POST new employee user account to /api/users	handleModalSubmit

Function Name	File	Start	End	Purpose	Called By
submitProfile	script.js	1892	1916	Validate and PUT current user profile update to /api/users/profile	handleModalSubmit
showToast	script.js	1922	1932	Display ephemeral toast notification badge in bottom-right stack	all API feedback handlers

5. IMPORTANT HTML ELEMENT IDS & CSS CLASSES

The JavaScript logic in script.js binds to specific semantic element IDs in index.html to manage application state and interactivity:

Element ID / Class	Type	Target Section	JavaScript Operational Purpose
#sidebar, #sidebar-backdrop	DOM Container	Navigation Shell	Drawer container toggled via toggleSidebar() and closeSidebar()
#menu-toggle	Button	Header	Triggers responsive sidebar opening on mobile viewports (<760px)
#page-title, #page-subtitle	Text Elements	Header	Dynamically updated with tab titles from pageCopy configuration
#global-search, #search-results	Input / Dropdown	Header	Triggers runGlobalSearch() live auto-complete across fleet records
#notify-btn, #notify-panel	Button / Dropdown	Header	Displays alert badge count and overdue ticket summary list
#user-btn, #user-menu	Button / Dropdown	Header	User identity chip opening Profile, Settings and Logout menu
#kpi-grid, #fuel-chart, #status-donut	Chart Containers	Dashboard	Populated dynamically with aggregated metrics and bar/donut SVGs
#vehicle-table-body, #vehicle-search	Table / Input	Vehicles	Populated with vehicles; filters table rows via renderVehicles()
#driver-table-body, #driver-search	Table / Input	Drivers	Displays driver roster; chip filters trigger status filter updates
#fuel-table-body, #fuel-vehicle-filter	Table / Select	Fuel	Renders fuel fill-ups and dynamic vehicle filter dropdown
#maintenance-board	Grid Container	Maintenance	Renders kanban columns; click handler marks tickets complete
#route-map, #route-table-body	Map / Table	Routes & Trips	Leaflet interactive road map and click-to-trace trip dispatcher
#user-table-body, #user-search	Table / Input	Users (Admin)	Renders tenant employees with instant account activation toggles
#modal-overlay, #modal-title, #modal-body	Modal Containers	Dialog Shell	Overlays for dynamic form injection (add-vehicle, add-driver, etc.)
#toast-stack	Fixed Container	Global UI	Container where showToast() dynamically appends temporary alert cards

6. FRONTEND EXECUTION FLOW

1. Application Bootstrap: When DOMContentLoaded triggers, checkAuthAndInit() issues a credentials-included fetch to /api/auth/me. If unauthenticated, the browser is redirected to /login.html.

2. Role Configuration: The user profile role (ADMIN or EMPLOYEE) determines DOM styling via document.body.classList.add("role-employee") and configures localized navigation labels.

3. Event Binding: bindEvents() registers click listeners on .nav-item tabs, filter inputs, modal submit buttons, search boxes, and dropdown toggles. All listeners are bound exactly once.

4. Parallel Data Loading: loadAllFromBackend() executes Promise.all across 7+ REST endpoints (/api/vehicles, /api/trips, /api/fuel, etc.). Admin sessions fetch driver and user rosters; Employee sessions retrieve strictly assigned records.

5. View Dispatching: renderAll() renders the active view. Tab switching via showSection(name) updates visibility classes without page reload.

6. Interactive Road Routing: When the Routes tab is opened, initOrUpdateRouteMap() initializes Leaflet with OpenStreetMap tiles. Clicking any trip fetches road geometry from /api/trips/:id/route (OSRM) and draws an accurate highway polyline.

7. FRONTEND DEVELOPER CHANGE GUIDE

Modification Goal	Primary File	Function / Section to Edit	Instructions & Guidelines
Change Dashboard KPI cards	script.js	renderDashboard() (Line 589)	Modify kpiCard() template strings or threshold calculations.
Add a column to Vehicle table	index.html & script.js	index.html (#page-vehicles) script.js: renderVehicles()	Add <th> in index.html and corresponding <td> template in renderVehicles().
Add a field to Add Vehicle modal	script.js	vehicleFormHtml() & submitVehicle()	Add field() call in vehicleFormHtml(); extract value in submitVehicle() payload.
Change button action or route	script.js	bindEvents() (Line 232)	Attach or update event listener targeting the relevant element ID.
Add navigation tab	index.html & script.js	index.html (<aside>) script.js: pageCopy & showSection()	Add <button class="nav-item"> with data-section attribute; register copy in pageCopy.
Update API fetch request	script.js	loadAllFromBackend() (Line 167)	Adjust endpoint URL or promise handler in Promise.all array.
Customize validation toast message	script.js	submit*() functions & showToast()	Update message string passed to showToast(msg, "error").

8. VISUAL REGRESSION VERIFICATION RESULTS

Visual inspection was conducted using automated browser execution across all primary authenticated views at 1536x776 resolution:

• Login Page (login.html): Split-screen dark navy left panel and clean rectangular form verified with zero badge clutter.

• Dashboard (index.html): Total Fleet (4), Active on Route (2), Maintenance Alerts (1), Monthly Fuel Spend (¹1,56,800) verified.

• Vehicles Tab: Table rows FL-101 through FL-104 rendered with proper status badges and View modal triggers.

• Drivers Tab: 4 drivers displayed with license numbers, safety scores, and Reassign buttons.

• Fuel Tab: Odometer progression and efficiency ratings (Above Average / Below Average) verified.

• Maintenance Board: Kanban columns Overdue (1), Upcoming (1), Completed (1) verified.

• Routes & Interactive Map: Leaflet interactive container with OSRM road geometry verified; click-to-trace operational.

• Route History Stacking Fix: Verified page scroll (350px down) clips map cleanly beneath the sticky topbar (z-index: 30). Topbar and user profile dropdown remain visually above map with zero bleed-through or tile overlap.

• Profile Dropdown Scrolled State: Opened profile menu while scrolled; verified menu displays completely above map with zero clipping.

• Browser Console: Inspected console logs during full navigation sequence; verified 0 errors.